LeetCode Website Analysis

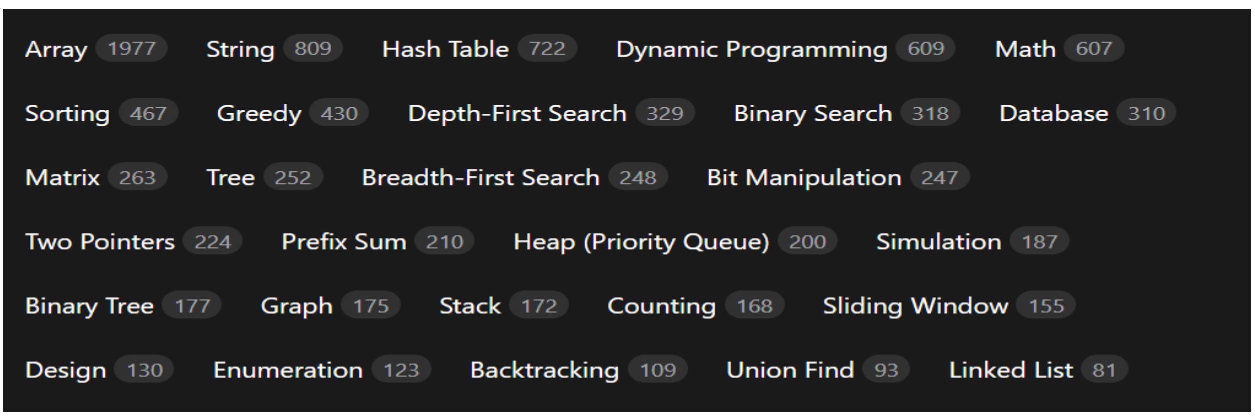
Overview

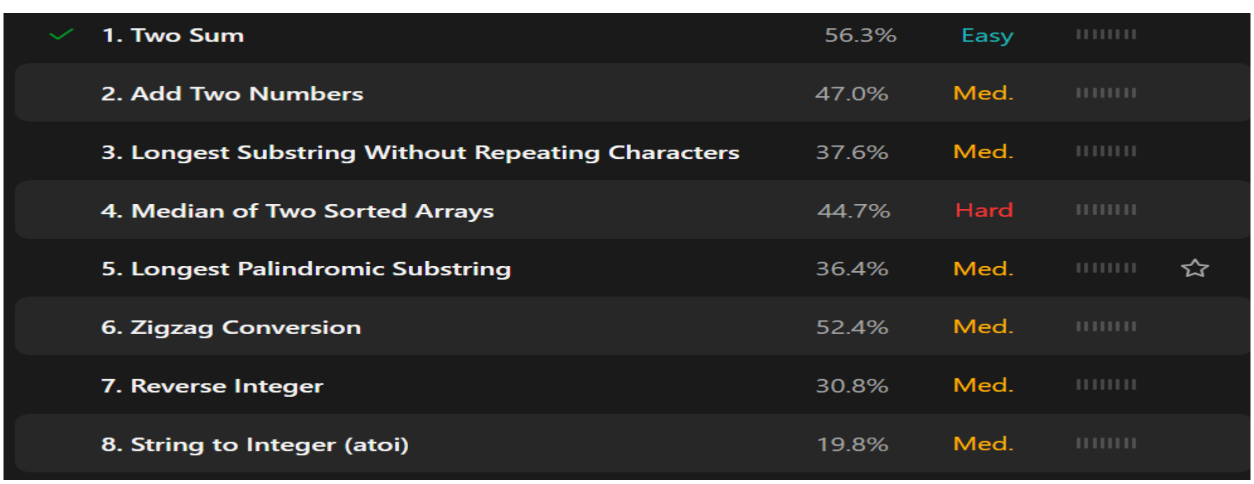
LeetCode is a platform focused on practice problems for coding interviews and algorithm learning. It provides a combination of problem-solving, contests, and premium content for interview preparation. Unlike Codeforces, which is more contest-focused, LeetCode balances practice + contest modes with guided learning paths.

User panel

User must register and login to use Leetcode.
Problem Set & Difficulty

- Problem Catalog: LeetCode has thousands of problems, categorized by difficulty (Easy, Medium, Hard), topic (arrays, graphs, dynamic programming), and company tags (Google, Amazon, Facebook, etc.).
- Premium Problems: Some problems require a premium subscription, typically company-focused or advanced algorithm problems.





- Problem Details: Each problem has a description, input/output format, examples, hints,a discussion section, Editorial, solutions, companies, similar questions, problem tags, acceptance rate etc.

Description | Editorial | Solutions | Submissions

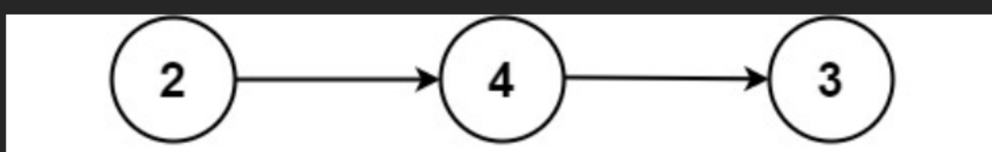
2. Add Two Numbers

Medium Topics Companies

You are given two **non-empty** linked lists representing two non-negative integers. The digits are stored in **reverse order**, and each of their nodes contains a single digit. Add the two numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

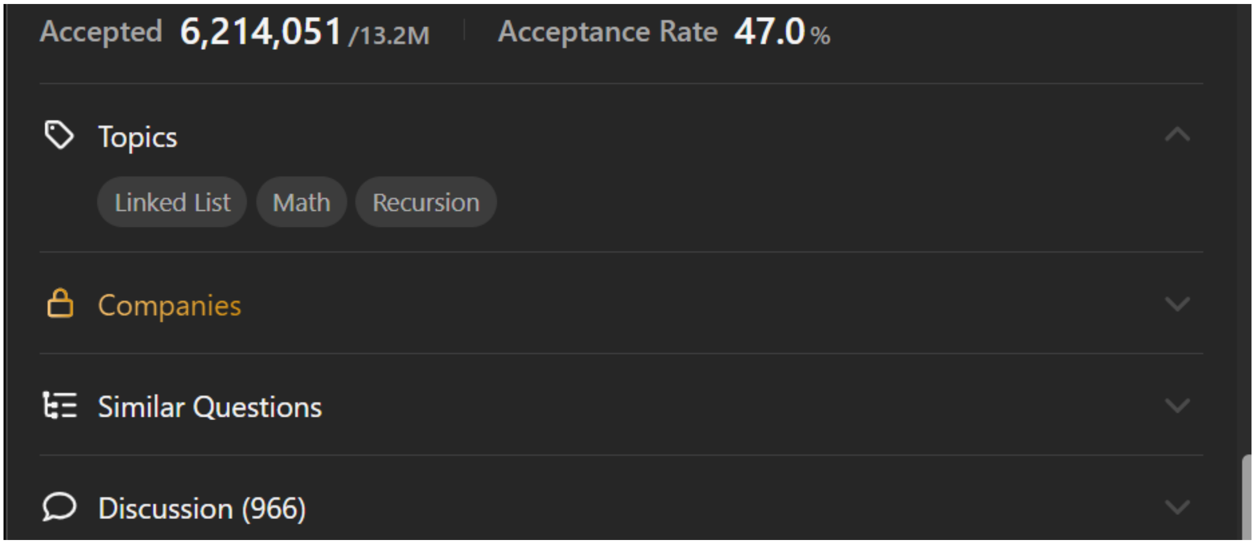
Example 1:



> **Input:** l1 = [2,4,3], l2 = [5,6,4]
> **Output:** [7,0,8]
> **Explanation:** 342 + 465 = 807.

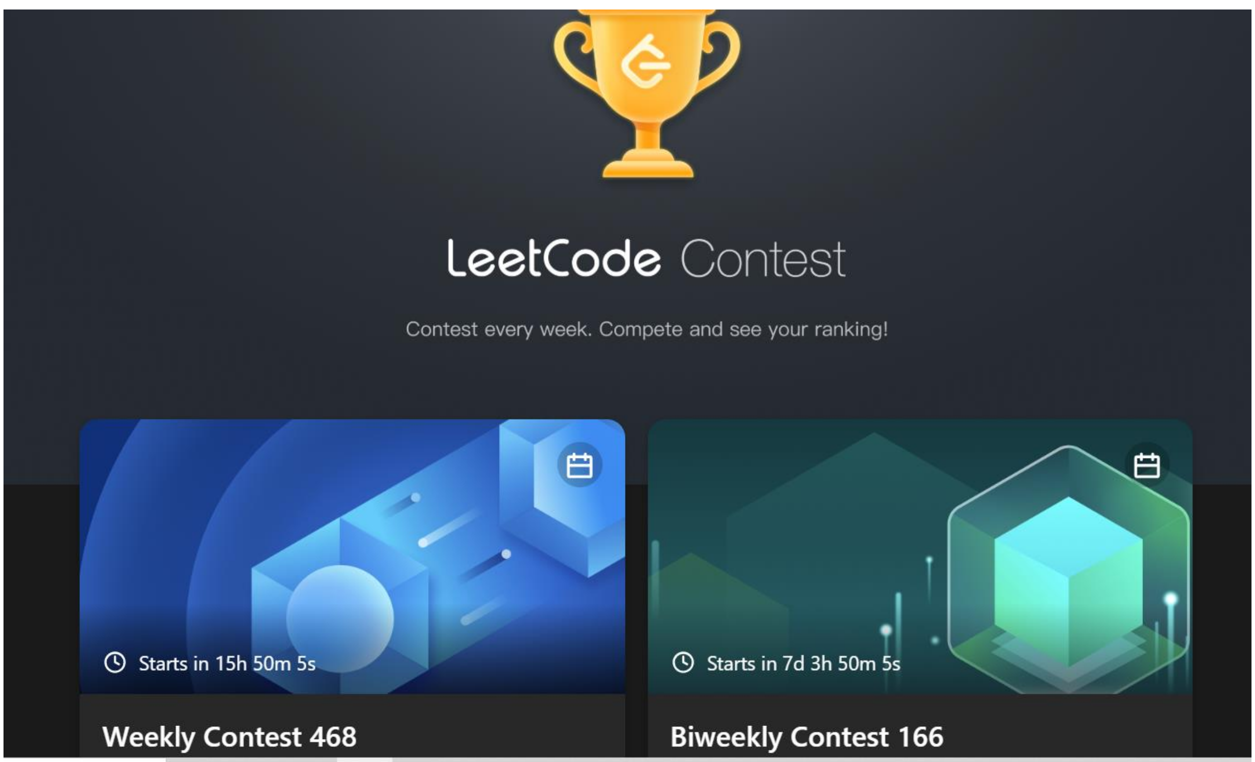
Example 2:

> **Input:** l1 = [0], l2 = [0]
> **Output:** [0]



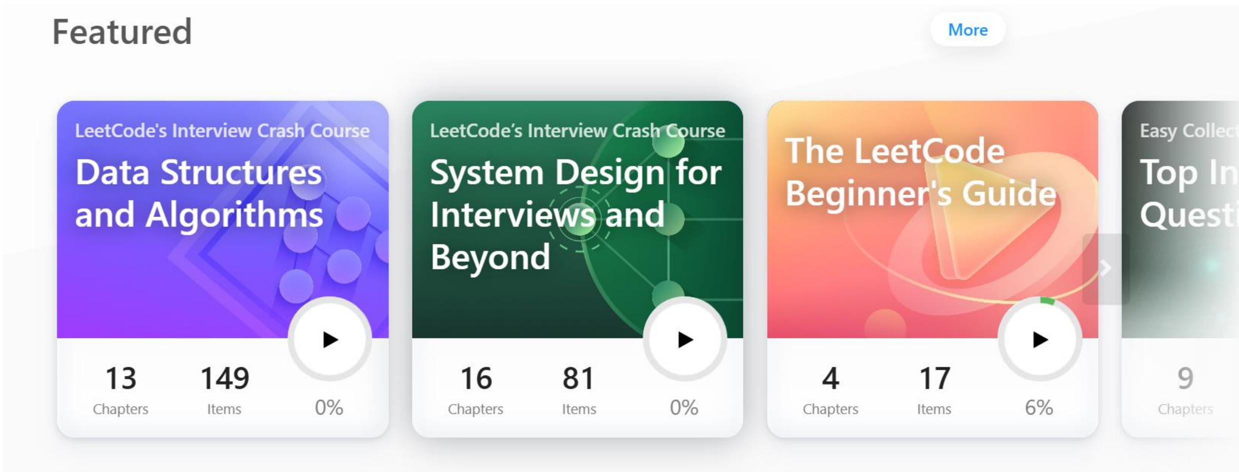
Contest System

- Weekly Contests:
    - Weekly LeetCode Contest, Biweekly Contest, and sometimes special contests.
    - Global leaderboard, ratings, and badges for top performers.
- Virtual Contests: Users can attempt previous contests at any time to practice under contest conditions.

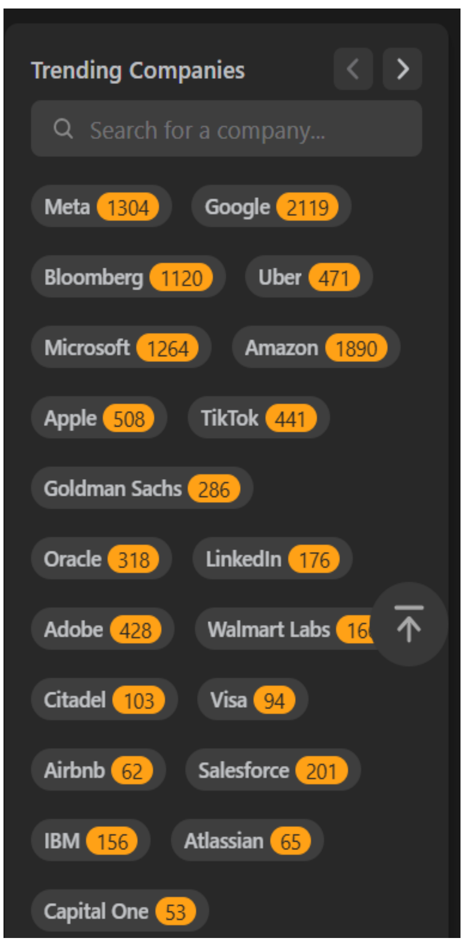


Learning & Guided Paths

- Explore & Learn Sections: LeetCode provides structured paths for learning algorithms, data structures, and interview skills.
- Mock Interviews: Simulate coding interviews with time limits and real-world questions.

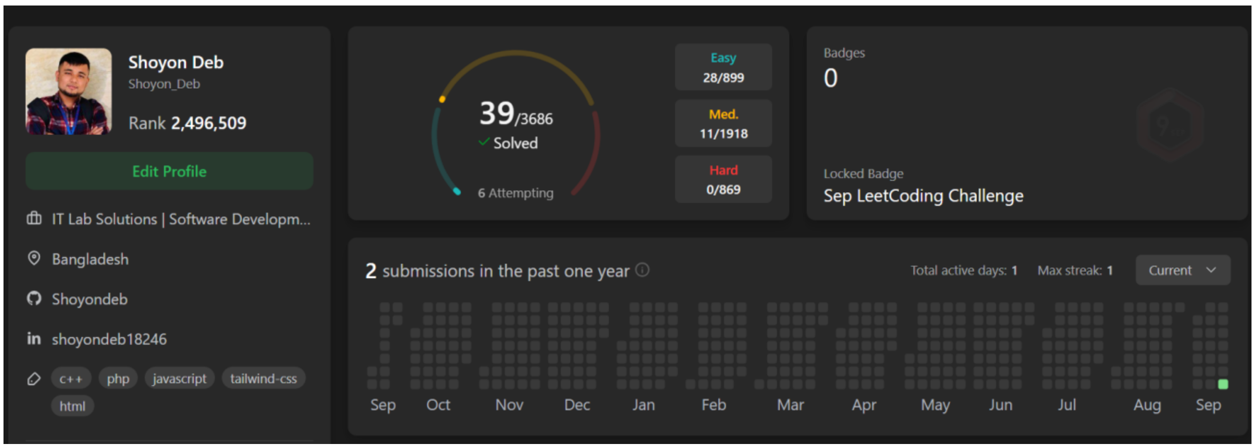


* Company-Specific Practice: Users can filter problems frequently asked in specific companies.



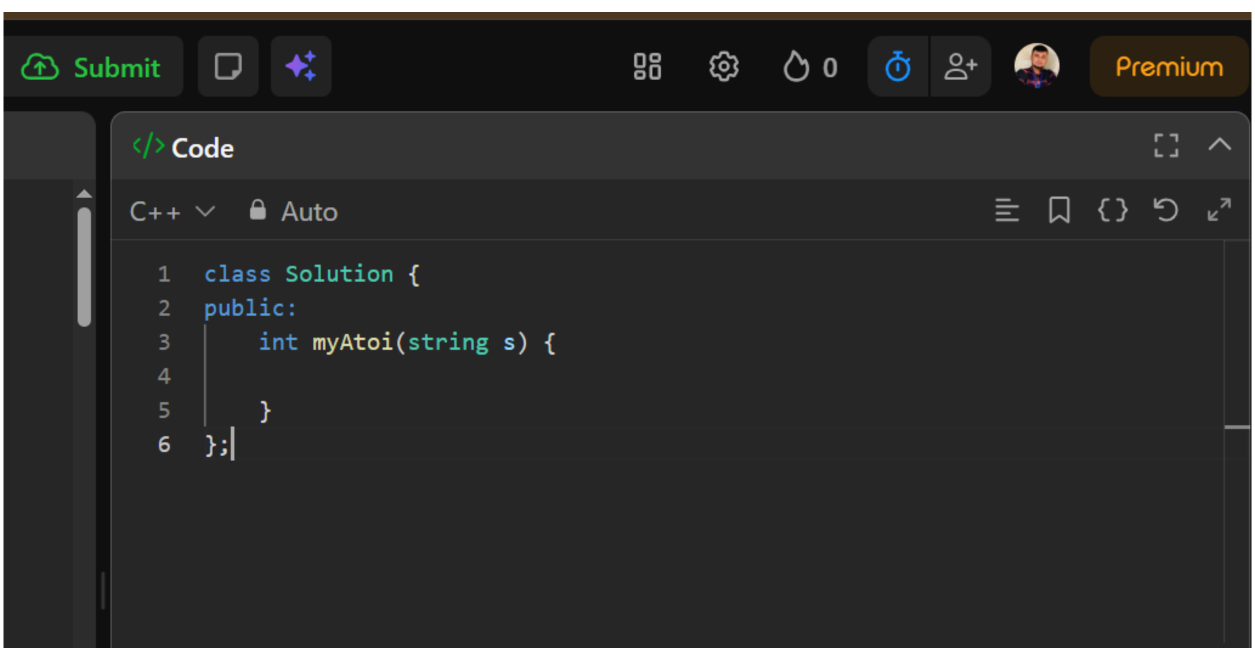
User Profiles and Analytics

- Profile Dashboard: Shows solved problems, contest ratings, badges, and achievements.
- Progress Tracking: Tracks solved problems by topic, streaks, submission history, and performance over time.
- Discussion & Hints: Users can view discussions, solutions, and hints for better learning.
- Leetcode encourage to manage everyday practice streak.



Code Editor & Submission

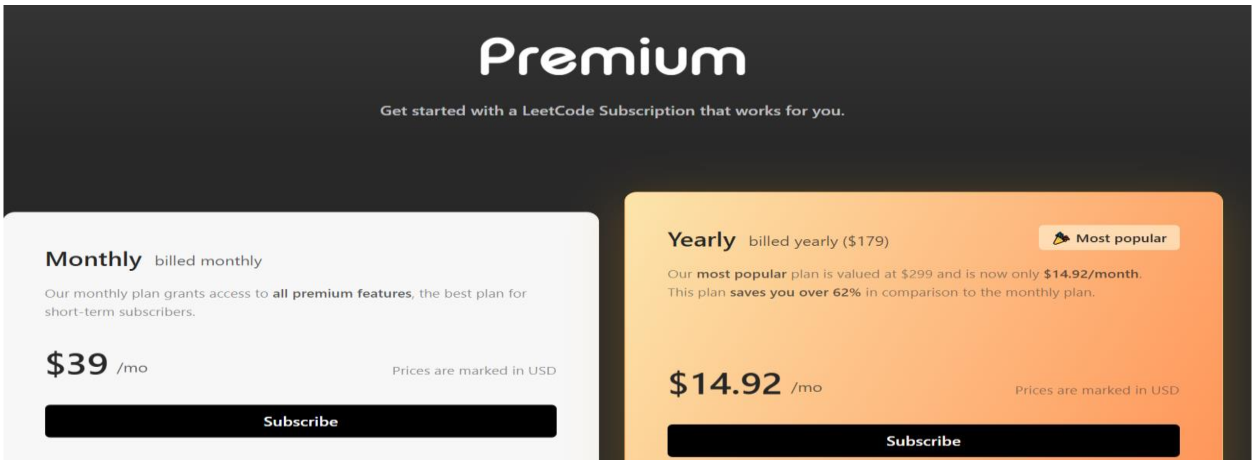
- Online IDE: Supports multiple languages (C++, Java, Python, JavaScript, etc.)
- Custom Test Cases: Users can run code against custom input for testing.
- There is a stopwatch and collaboration option.



- Immediate Feedback: Submission returns results (AC, WA, TLE, MLE, RE) with runtime and memory usage.

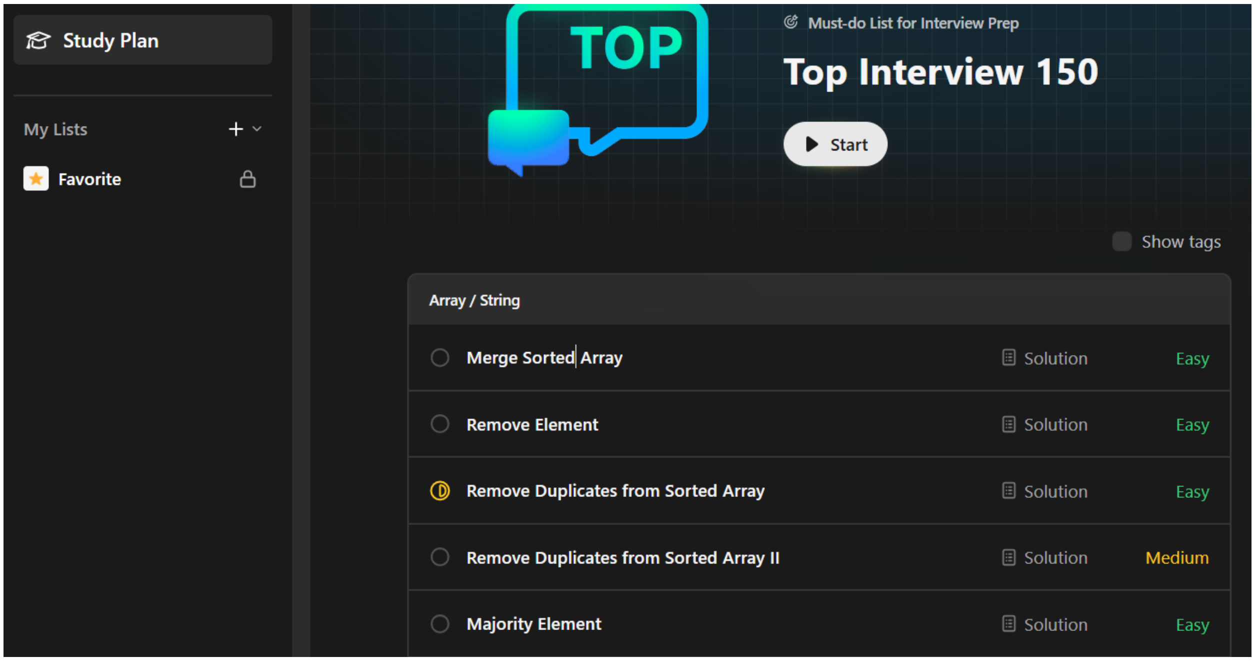
Premium Features

- LeetCode Premium Subscription includes:
    - Access to exclusive problems.
    - Company-specific interview questions.
    - Video explanations and premium solutions.



Leetcode suggests a study plan with some important questions:

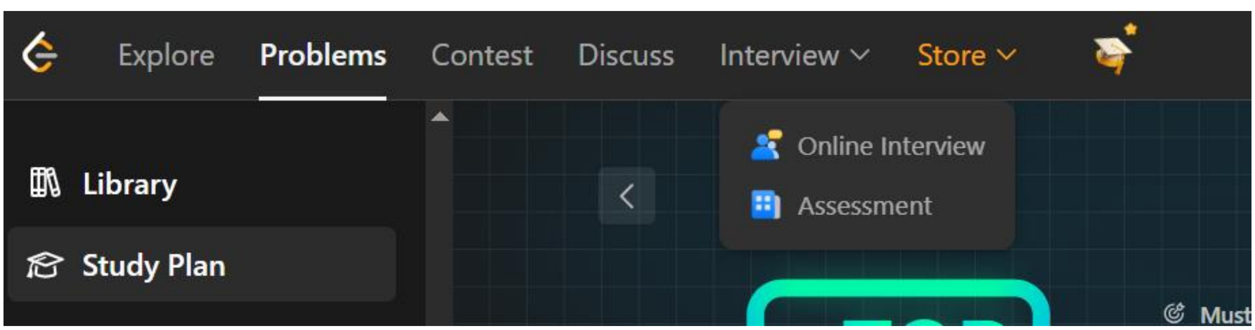
A list of questions can be created using my list section in the side bar.



User Interface and Experience

- Modern UI/UX: Clean, professional, and mobile-friendly interface.
- Navigation: Problems are easily searchable via tags, difficulty, or companies.
- Gamification: Badges, streaks, contest ratings, and leaderboard motivate users.
- Mobile App: Available for iOS and Android for coding on the go.

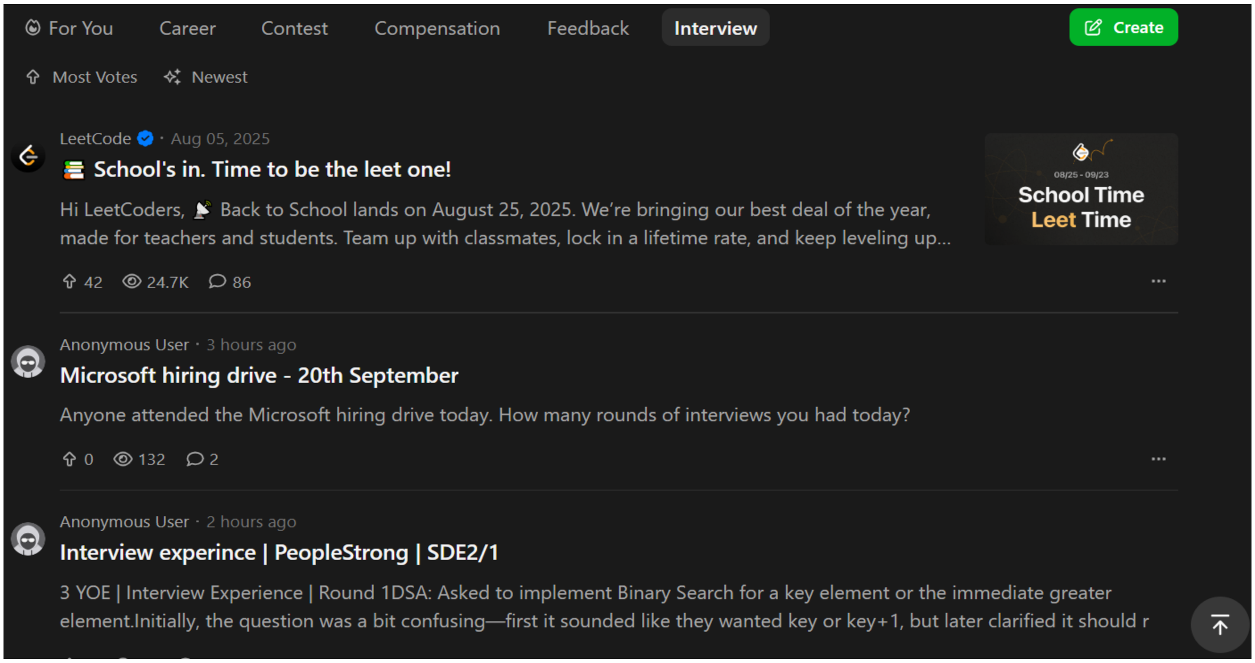
Leetcode offer free interview assessment test:



Community & Discussion

- Discussion Forums: Each problem has a discussion section where users can share solutions, hints, and optimizations.
- Editorials & Solutions: Official and community-written explanations help users learn approaches.

- Company Tags: Users can collaborate and share insights on problems often asked in specific companies.
- User can post his/her requirements on discussion section



Strengths

- Huge problem database with categorized difficulty and company relevance.
- Modern, clean, and mobile-friendly interface.
- Structured learning paths for interview preparation.
- Strong analytics on performance, streaks, and weak topics.
- Premium content for advanced learning and company-specific preparation.

Improvement Points

- Some premium problems are locked, limiting free users.
- Limited contest frequency compared to Codeforces (mostly weekly/biweekly).
- Gamification is less competitive; fewer social features than Codeforces.

Admin Panel Functionalities

The admin panel is used by LeetCode staff (admins, problem setters, content managers) to manage users, contests, and problems.

User Management

- View and manage user accounts.
- Handle account issues (password reset, bans for violations).
- Track suspicious behavior or plagiarism in submissions.

Problem Management

- Add new problems with description, constraints, input/output examples.
- Tag problems with difficulty, topic, and company relevance.
- Upload test cases for judging.
- Add editorials or hints for problems.

Contest Management

- Schedule weekly or biweekly contests.
- Configure contest type (single round, multi-day contest, virtual).
- Assign problems to contests.
- Set scoring rules (full score, partial score, time-based scoring).
- Monitor active contests in real time.

Analytics & Reporting

- Monitor submissions, success rates, and platform activity.
- Generate reports on contest participation and performance.
- Analyze trending problems and topics for future contests.

Platform Maintenance

- Announcements, notifications, or updates.
- Monitor system performance, uptime, and server health.
- Backup and restore databases if necessary.

Judge Panel

The judge panel is the automated system that evaluates all submissions (practice and contest).

Submission Intake

- Receive code submissions from users.
- Identify programming language.
- Queue the submission for evaluation.

Code Compilation

- Compile code in a sandboxed environment.
- Catch compilation errors before running.

Test Case Execution

- Run the code against hidden test cases (system + sample cases).
- Measure runtime, memory usage, and correctness.

Verdict Assignment

- Assign one of the following verdicts:
  - Accepted (AC)
  - Wrong Answer (WA)
  - Time Limit Exceeded (TLE)
  - Memory Limit Exceeded (MLE)
  - Runtime Error (RE)
  - Compilation Error (CE)

Scoring & Ranking

- In contests, calculate score based on problem weight, partial scoring, and submission timing.
- Update contest leaderboard in real time.
- Handle virtual contests where rating is not affected.

Performance & Monitoring

- Distribute load across multiple servers for large contests.
- Monitor judge queue, execution times, and system resources.
- Detect potential cheating or plagiarism using similarity analysis.